

Capital Region Planning Commission



Project Name: Delta Regional Authority (DRA) Local Development District (LCD)
Community Pilot Project Fund

Objective

- Support Distressed and underserved communities CRPC's 11-parish region by providing recommendation to government bodies

Task

Build a decision-support tool to visualize data that enables **CRPC officers** making **faster, more accurate, and more efficient decisions** with minimal processing time.

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CRPC Region – DRA Distress Status Visualization and AI Assistance

Project Report

Project Demo Video Link: [Capital Region Planning Commission CRPC Data Visualization and AI Assistance Internship Project - YouTube](#)

1. What This Project Does

This project is an interactive web-based mapping application that visualizes economic distress status for all 252 census tracts across the 11 parishes of the Capital Region Planning Commission (CRPC) area in Louisiana. It classifies each tract using DRA-style (Delta Regional Authority) distress criteria — per capita income and unemployment rate relative to the U.S. average — and layers on job and business-establishment data, plus a natural-language AI assistant for exploring all of it conversationally.

- **Distress mapping.** Color-coded tract map (Non-Distressed, Distressed by Unemployment, Distressed by Income, Distressed by Both, or No Data) built with Folium/Leaflet.
- **Interactive filtering.** Filter by parish, distress status, or by two adjustable range “knobs” (per capita income and unemployment rate) with a 10-bin thermal gradient scale; all filters combine together.
- **Job density overlay.** Tract- and parish-level job counts by NAICS sector from LODS workplace data, shown as a size/color-coded map overlay.

- **Business establishments overlay.** Parish-level business counts (total establishments, plus micro/small/medium/large employer breakdowns) from County Business Patterns data, shown the same way.
- **AI assistant.** A chat box answers natural-language questions (e.g. “show the 10 lowest income tracts in Ascension Parish” or “which parish has the most small businesses?”) using a cloud LLM, with conversation memory and curated example prompts.
- **Export and full screen.** A server-rendered 600 DPI map export (with an active-filters summary panel) and a full-screen map mode.

Screenshots of the running application, illustrating the features described above:

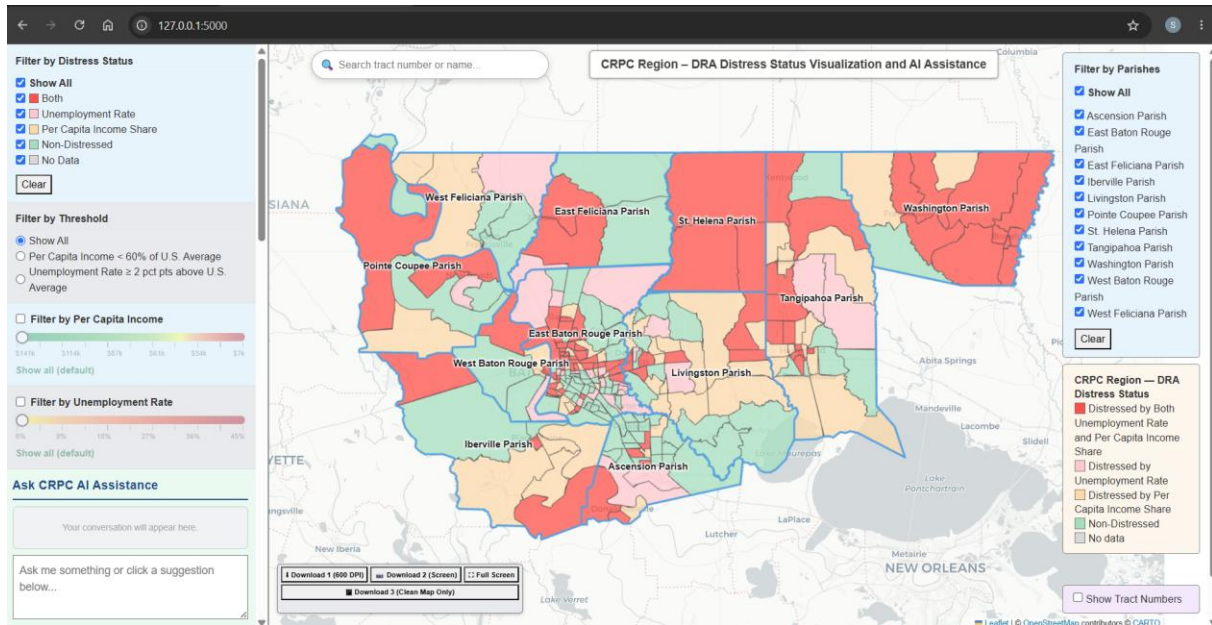


Figure 1. The main CRPC visualization interface (<http://127.0.0.1:5000>). The center pane is the interactive Leaflet/Folium map of all 252 census tracts across the 11 CRPC parishes, color-coded by DRA distress status (red = distressed by both income and unemployment, pink = distressed by unemployment only, tan = distressed by income only, green = non-distressed, gray = no data), with a tract/name search bar and page title above it. The left sidebar holds the “Filter by Distress Status” and “Filter by Threshold” panels plus the per-capita-income and unemployment-rate range-knob sliders, and the “Ask CRPC AI Assistance” chat box below them. The right sidebar holds the “Filter by Parishes” panel, the distress-status color legend, and a “Show Tract Numbers” toggle. The three buttons along the bottom (Download 1–600 DPI, Download 2–Screen, Download 3–Clean Map Only, plus Full Screen) export the current map view as an image.

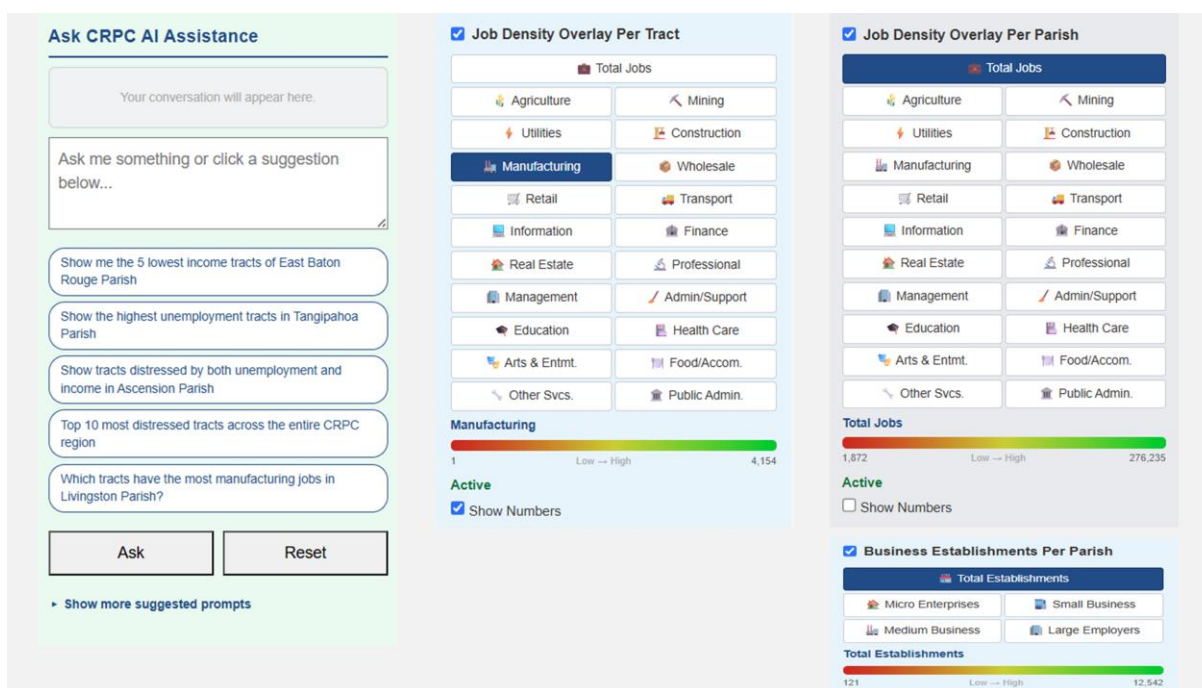


Figure 2. Detail views of the remaining tool panels. Left: the “Ask CRPC AI Assistance” chat panel, with its conversation window, free-text question box, Ask/Reset buttons, and a set of clickable example prompts (e.g. lowest-income tracts, highest-unemployment tracts, tracts distressed by both criteria, most-distressed tracts region-wide, tracts with the most jobs in a given sector). Middle: the “Job Density Overlay Per Tract” panel, a grid of buttons for Total Jobs plus 20 individual NAICS employment sectors (Agriculture, Manufacturing, Retail, Health Care, etc.); selecting one color-codes every tract by that sector’s job count using the low-to-high gradient legend shown, with an optional “Show Numbers” label overlay. Right: the same sector grid aggregated to the parish level (“Job Density Overlay Per Parish”), stacked above the “Business Establishments Per Parish” panel, which color-codes parishes by total business establishments or by micro/small/medium/large employer counts from County Business Patterns data.

2. Data Sources Used

All datasets are pre-processed extracts limited to the 11 CRPC parishes (Ascension, East Baton Rouge, East Feliciana, Iberville, Livingston, Pointe Coupee, St. Helena, Tangipahoa, Washington, West Baton Rouge, and West Feliciana), Louisiana.

Source	Description	Used For
U.S. Census Bureau Cartographic Boundary Files (2025, 500k)	Census tract polygon geometries (TIGER/Line cartographic boundaries) for Louisiana.	Tract shapes drawn on the map.
FFIEC Census Tract List (2026)	Federal Financial Institutions Examination Council tract list — tract GEOIDs and geography names for the CRPC parishes.	Matches distress labels to the correct tract geometry.
Distress-Labeled Tract Data (derived, per parish)	Per capita income and unemployment rate (5-Year ACS) with DRA-style threshold calculations, labeled Yes/No for each distress category per tract.	Core distress classification and color-coding.
LODES WAC (2023)	LEHD Origin-Destination Employment Statistics, Workplace Area Characteristics — block-level job counts by 20 NAICS sectors, aggregated to tract level.	Job density overlay and AI job-related queries.

County Business Patterns (2023)	U.S. Census Bureau establishment counts by employer size class, filtered to Louisiana and the 11 CRPC parishes.	Business establishments overlay and AI business queries.
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The diagram below maps each dataset used by the tool back to its original source:

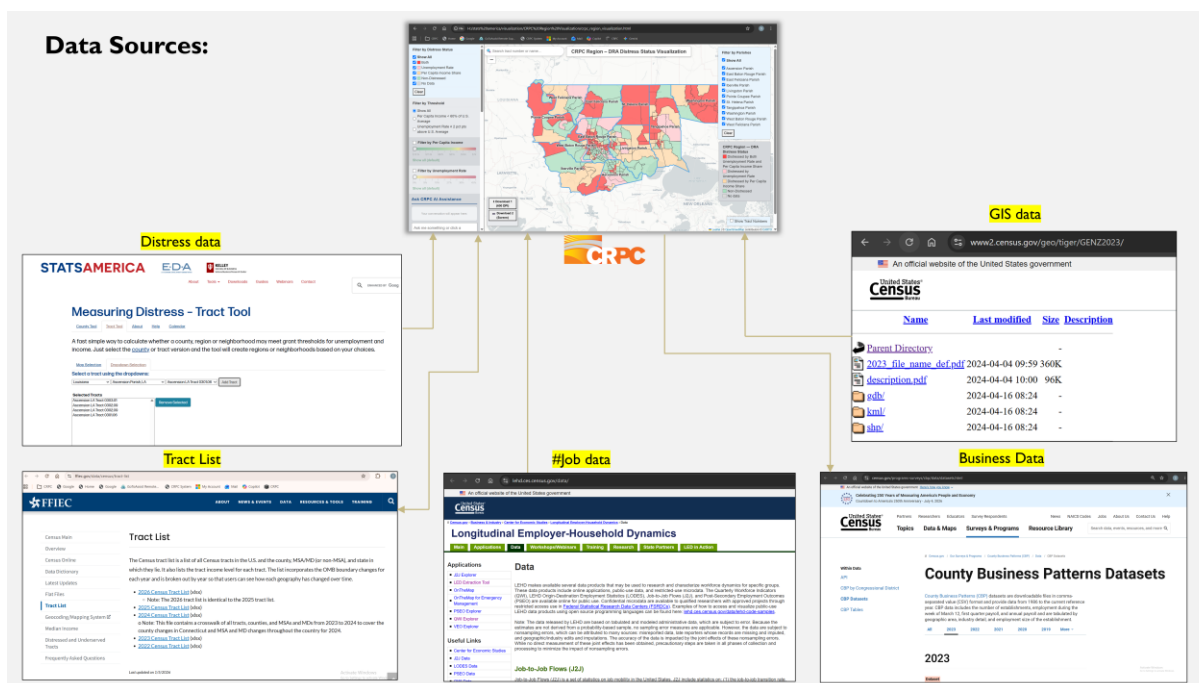


Figure 3. Data source map for the CRPC visualization tool. The central screenshot is the tool itself; each arrow traces one of its inputs back to the government or research website it was pulled from. “Distress data” — StatsAmerica’s “Measuring Distress – Tract Tool” (EDA / Kelley School of Business), used to compute the DRA-style per-capita-income and unemployment-rate distress thresholds by tract. “Tract List” — the FIEC census tract list, the official Census-tract-to-county/MSA crosswalk used to enumerate and align tracts across years. “#Job data” — the Census Bureau’s LEHD (Longitudinal Employer-Household Dynamics) data page, source of the LODES workplace employment counts behind the job density overlay. “GIS data” — the Census Bureau’s TIGER/Line boundary files, providing the tract and parish shapefiles the map is drawn from. “Business Data” — the Census Bureau’s County Business Patterns (CBP) datasets, source of the establishment counts behind the business establishments overlay.

3. System Setup Needed to Run

- **Operating system.** Windows 10/11 (the app auto-registers native GDAL DLLs on startup for Windows).
- **Python environment.** Conda/mamba (e.g. Miniforge) with a Python 3.11 environment named “crpc”, created from the included environment.yml (mamba env create -f environment.yml). A requirements.txt is provided as a pip fallback, though the geospatial packages (GDAL/GEOS/PROJ via geopandas, rasterio, contextily) install far more reliably through conda/mamba on Windows.
- **Key packages.** Flask, folium, geopandas, pandas, matplotlib, contextily, rasterio, shapely, pyproj, openai, and python-dotenv (all pinned in environment.yml).
- **AI assistant key.** A .env file with a GROQ_API_KEY (or LLM_API_KEY / LLM_BASE_URL / LLM_MODEL for another OpenAI-compatible provider) enables the AI assistant chat box. Everything else — map, filters, overlays, download — works without it.

- **Internet connection.** Required for the basemap tiles (CartoDB Positron) and for the AI assistant's calls to the LLM API. All tract, job, and business data itself is stored locally and works offline.
- **Running the app.** Run "python visualize_crpc_region_with_filter22.py" from the crpc environment. It starts a local Flask server; once "Running on http://127.0.0.1:5000" appears, open that address in a browser. It is a long-running server (stop with Ctrl+C), and only one instance can run at a time on port 5000.

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